SEMESTER PROJECT PART 2 - ITERATION 1

Working first version, implementation review, and follow-up archive feature

Student Shubham	**UNI** SL5836
Chosen repository funnystudy	**Report date** November 6, 2026
Assignment Build and review your first version	**Review status** PM feedback pending

Executive summary

This project improves the participant journey in the Humor Study app. The first version brings setup progress together, clarifies whether a study is new, in progress, or completed, saves a resumable study checkpoint, and shows saved completion results. A subsequent requested enhancement adds a read-only archive for ended public studies so participants can browse old captions without reopening participation or submitting votes.

Submission integrity: the implementation and automated verification are complete, but the PM demonstration and feedback round have not yet been recorded. This report does not invent PM feedback or claim approval.

At a glance

Problem addressed Unclear setup and return-to-study state	**Primary user** Nadia, composite adult research participant
Core outcome Resume at saved study position	**Follow-up** Safe browsing of ended public studies

1. User, problem, and proposed direction

User persona

Nadia is a fictional composite adult research participant, not the project owner. She is willing to participate in a humor study but can be interrupted, may not remember which setup steps are complete, and may return later after a break. She needs to know what remains and to continue without losing her place.

Problem statement

Participants pass through multiple steps before starting: sign-in, notification choices, demographics, profile completion, and then the study itself. The prior experience was functional but could feel fragmented. It did not consistently summarize setup state or make the consequences of returning to an unfinished or completed study obvious. Losing context after interruption adds friction and may reduce completion.

Proposed direction and expected outcome

The proposal focuses on visible onboarding and study progress, clear Start / Resume / View Results actions, a reliable continue flow, and a completion summary. The intended outcome is a more understandable and recoverable participant journey, not an unrelated feature added without a user need.

How the work supports multiple feedback iterations

- **Iteration 1:** build the end-to-end status and resume foundation, then demonstrate it to the PM.
- **Iteration 2:** use PM observations to refine labels, checkpoint visibility, and return behavior.
- **Iteration 3:** validate interruption/re-entry and completion messaging; prioritize any remaining usability issue from feedback.

A later user-requested extension adds a distinct read-only history use case: browsing previous studies when no study is currently active. It remains separate from the participation/resume flow.

2. What is implemented

Participant setup dashboard

- A home-page checklist summarizes Profile, Notifications, and Demographics.
- Each step displays a state, a short explanation, and a Continue destination when action is needed.
- Study start/resume is disabled until required profile and onboarding steps are complete.
- The responsive cards stack their status below each heading, use adaptive columns, and avoid text overflow; the profile heading uses a theme-aware foreground color for readable contrast.

Study lifecycle and saved progress

- Study cards distinguish Start study, Resume study, and View results; an in-progress card shows the saved caption position where available.
- A validated database checkpoint stores the stable caption order, current caption cursor, and run-level viewed/liked/timing data.
- On return, the study loads around the saved cursor and restores the run state instead of beginning at the first caption.
- Completion persists an immutable stats snapshot; completed studies reopen the same results, including after the active window has expired.

3. Participant interface and behavior

Dashboard evidence

The image below is a screenshot shared by the user of the participant dashboard. It shows the 1-of-3 setup summary and the no-active-studies state. It was captured before the later archive link was added; the current empty state now also offers Browse previous studies.

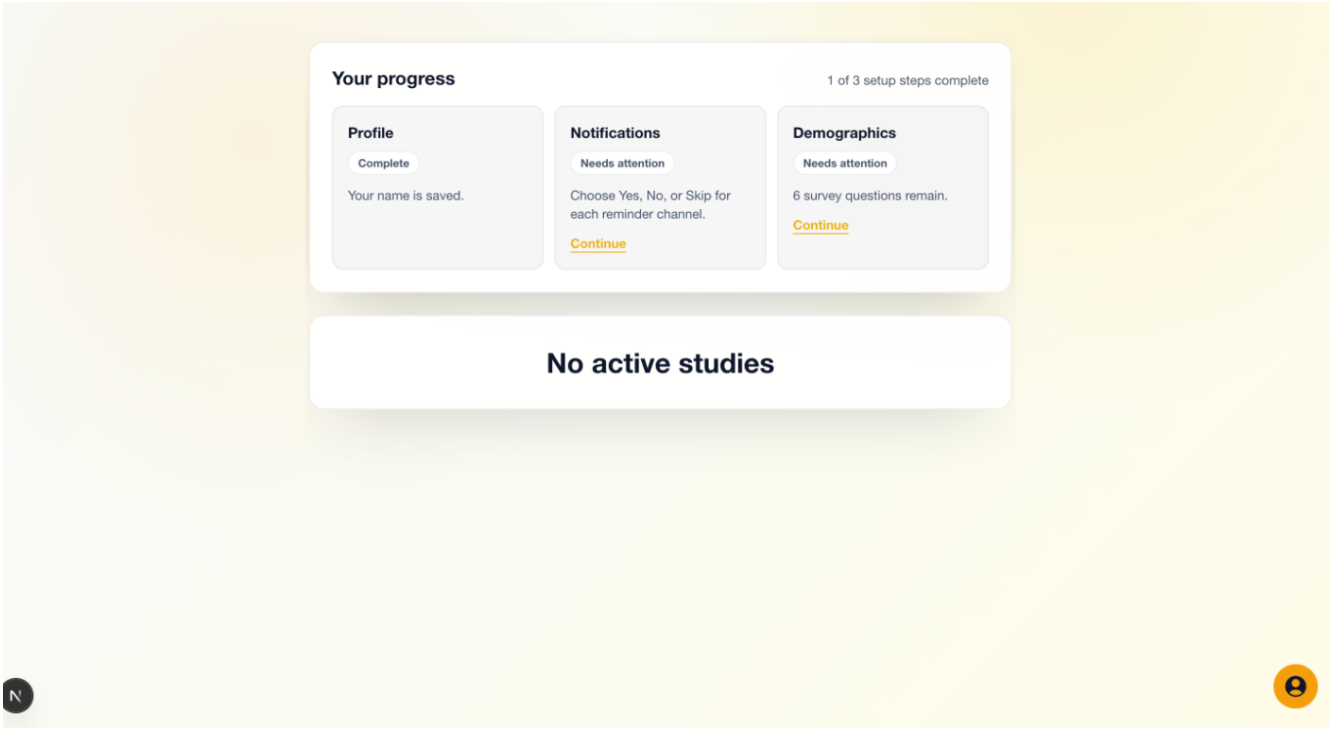


Figure 1. User-provided dashboard screenshot (earlier state; no active studies).

Study status and result states

The following is an illustrative UI preview, not a screenshot of a live account. It shows the three implemented home-card actions. Sample study IDs and progress are synthetic and are not participant records.

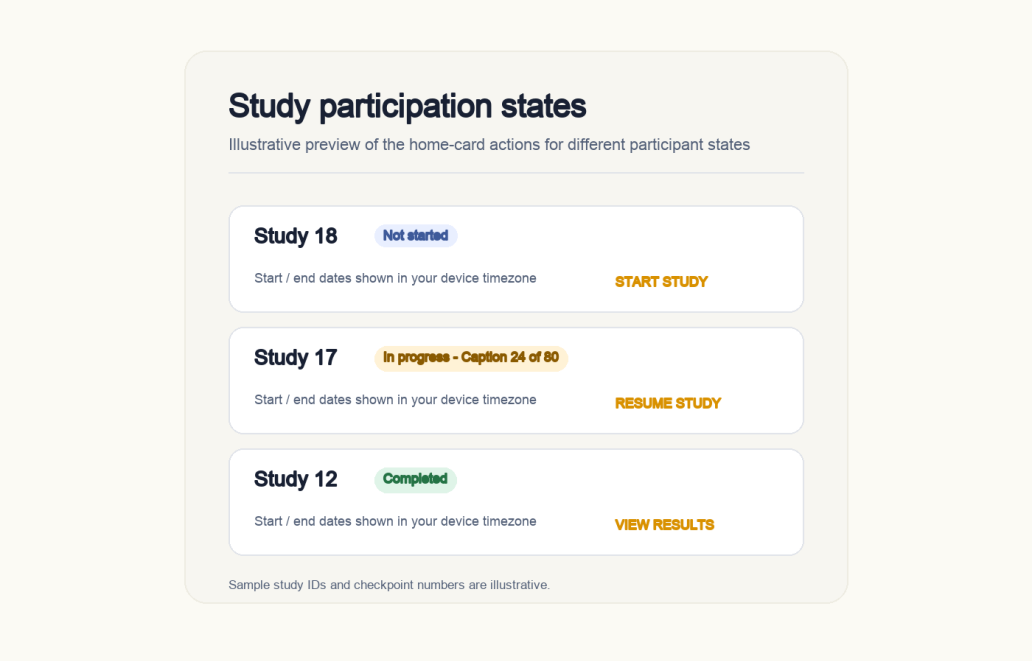


Figure 2. Illustrative preview of Start, Resume, and View Results states.

4. Resume and completion experience

Resume behavior

When a participant begins, the app saves progress through the authenticated user's Supabase session. Caption order is stored for that run, so resuming does not reshuffle the participant into a different sequence. The initial fetch is centered around the saved position while subsequent caption details continue loading in batches. The progress payload is validated in the application and database.

Completion behavior

Ending a study saves the final participation time and summary together. The results screen reports median time before liking, median time before the next caption, upvote percentage, captions liked, and captions viewed. It reads the saved completion snapshot rather than fabricating stats or depending on captions remaining active.

The results illustration uses sample values only; actual results are participant-specific.

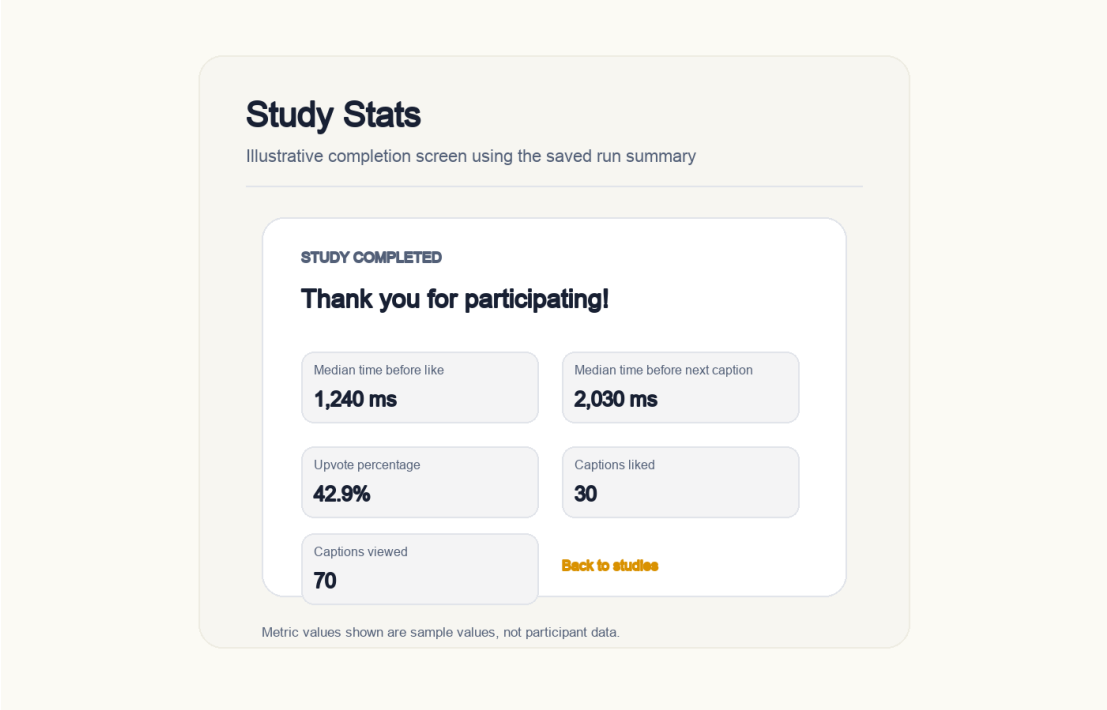


Figure 3. Illustrative completion-summary preview with sample metrics.

Data and authorization safeguards

- Progress save operations use the signed-in participant's JWT; no service-role key is used by this feature.
- The database validates checkpoint structure and grants narrow participant update access to progress and completion columns.
- Completed snapshots are immutable on retry; backward navigation does not end participation.

5. Follow-up feature: previous-study archive

The home page now links to **Browse previous studies**, including when there are no active studies. The archive lists ended public studies and opens a paginated, read-only caption gallery. It is intentionally not a way to restart an expired study.

Archive constraints

- Only studies marked public and with an end time in the past are listed.
- Private, upcoming, still-active, and missing studies are not shown in the archive viewer.
- The viewer reads caption mappings in bounded pages (24 captions per page) to avoid fetching an entire study's content at once.
- Archive UI contains no vote, bookmark, start, end, or progress-save controls.
- The archive does not create participation rows or call study write actions; normal study dates and participation rules remain unchanged.

Because the browser was signed out during report creation, the archive screenshots below are explicitly illustrative UI previews built with sample labels and records. They do not represent a live authenticated session or actual database contents.

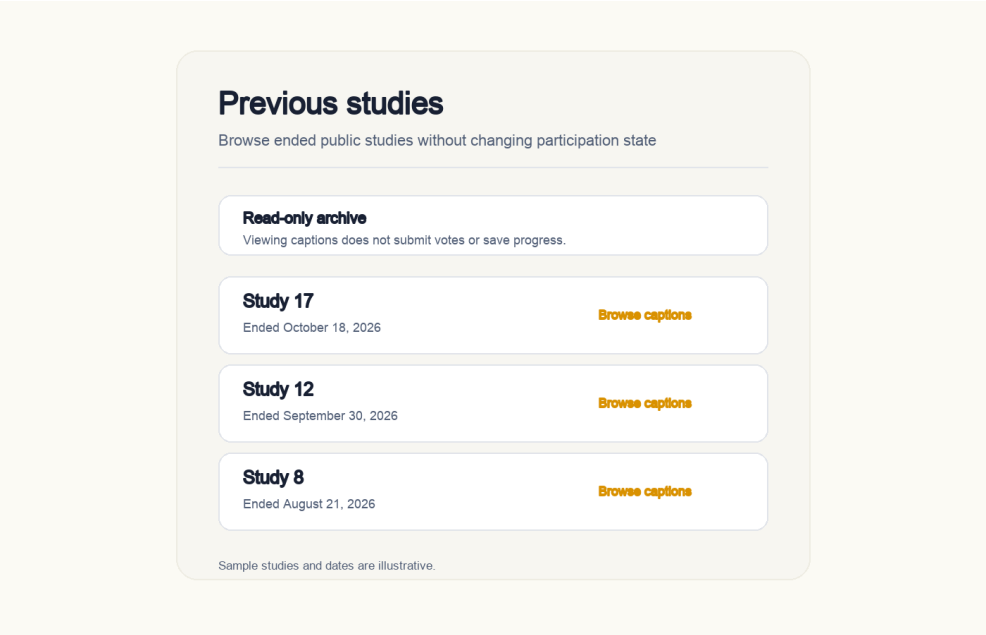


Figure 4. Illustrative previous-studies list; displayed records are sample content.

6. Archive browsing and implementation

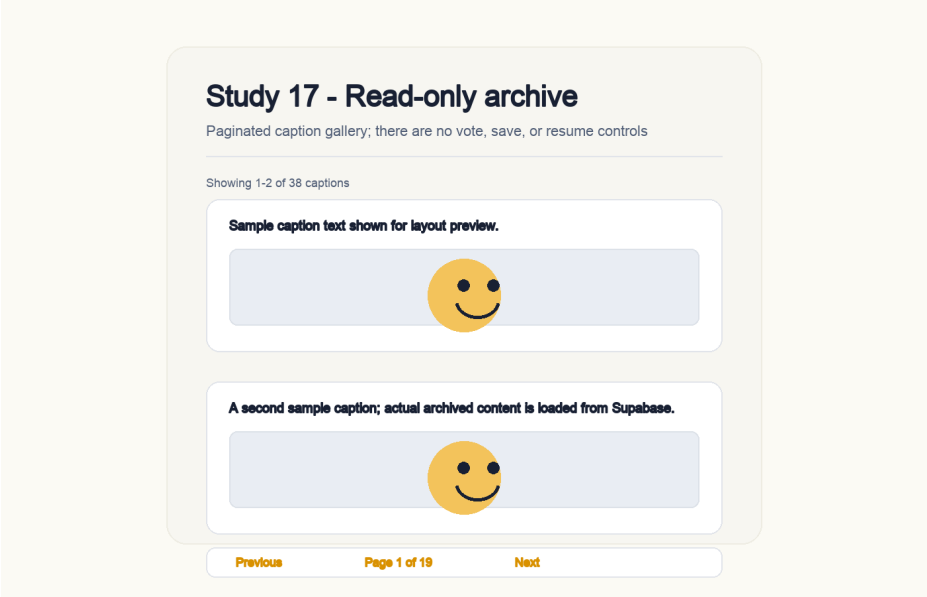


Figure 5. Illustrative paginated read-only caption gallery; sample captions/images are not live data.

Implementation map

Area	Implementation	Purpose
Home/dashboard	Progress checklist, study state cards, archive entry point	Orient participants and guide next action
Study flow	Saved cursor, stable order, timing/view stats	Recover from interruption and preserve run context
Database	study_participations.progress, validation, authenticated save function	Persist and constrain resume data
Archive routes	/previous-studies and /previous-studies/{studyId}	Browse ended public study captions without participation writes

Repository areas touched

The implementation spans the Next.js home and numeric study routes, reusable participant progress and completion components, study loading and progress helpers, authenticated study actions, Supabase database types and migration, regression tests, and the project/iteration documentation.

Representative files: `src/app/page.tsx`, `src/app/previous-studies/page.tsx`, `src/app/previous-studies/[studyId]/page.tsx`, `src/app/study/[studyId]/page.tsx`, `src/components/ParticipantProgress.tsx`, `src/components/CaptionVoteFlow.tsx`, `src/components/CompletedStudy.tsx`, `src/lib/studies.ts`, `src/lib/study-progress.ts`, `supabase/migrations/20261105153000_add_resumable_study_progress.sql`, and `tests/demographic-routing.test.mjs`.

7. Data setup and verification

Teaching-data environment

The anonymized course teaching-data dump was restored into the newly configured Supabase project after verifying it was empty. The restore verifier reported 95 tables, 836,827 rows, 261 foreign keys, and 90 application policies. The resumable progress migration was applied and the rollback-only participation SQL regression test passed. No database passwords, API keys, or OAuth client secrets are included in this report.

Application checks

Check	Result	Coverage
npm run test:demographics	37 passed, 0 failed (after archive feature)	Study routing, onboarding, dashboard states, resume validation, archive filtering/pagination
Focused ESLint	Passed	Changed app routes, study loaders, and regression tests
npm run build	Passed	Production compilation, TypeScript, and route generation
Database SQL regression	Passed earlier; rollback-only	Participation timestamps, checkpoint validation, permissions, isolation

Review status, limitations, and next steps

PM review and feedback are pending. The working version must still be demonstrated to the PM. The assignment asks for feedback on usefulness, clarity, and behavior; these are left as questions rather than represented as completed review.

- Demonstrate the checklist, study start/resume flow, interruption recovery, completed stats, and archive to the PM.
- Record actual feedback and any observed confusion; prioritize the highest-impact change for the next iteration.
- Test on a signed-in account in the configured project; the report's archive and state previews use illustrative sample content.
- Ask the PM whether the history archive is useful, whether read-only behavior is clear, and whether caption page size/navigation feels appropriate.

This document describes the working local implementation and verification performed. Deployment to a hosted environment and PM acceptance are separate steps.